

Paul Abrams

Ottawa, Ontario

.NET / C# / Blazor Developer · Reusable Component UIs · Azure DevOps Pipelines & Work-Item Delivery · 15+ Years Government of Canada Web Portals · Professional Scrum Master

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Professional Summary

Hands-on .NET developer with 15+ years building and improving **web-based portals** across the Government of Canada — **Blazor / C# / .NET** front and back end on **cloud-native, microservices** architectures. **Directly relevant to the City of Ottawa engagement:** I have built and enhanced **resident- and business-facing portals** (a Blazor onboarding portal at ESDC; the “**My iCase**” **personalization portal** used by 5,000+ lawyers and external agent firms), expanded **personalization functionality** (a customizable, widget-based portal backed by a **custom personalization provider**), and shipped **notification functionality** (event-driven **Azure Functions**, scheduled digest emailers, and availability alerting). I **translate designs into reusable, component-based UIs**, and I am **deeply fluent in Azure DevOps** — analyzing and completing **work items** against sprint timelines, collaborating with Product Owners on acceptance criteria, and **leveraging ADO pipelines to automate build, test, and deployment**. I work the way the role describes: a **simplified SCRUM** cadence of daily stand-ups, sprint planning, and sprint reviews, with **documentation and active knowledge-sharing** throughout. **Security- and reliability-minded by default** — IAM via **OAuth 2.0 / OpenID Connect / Entra**, migration to **Managed Identities**, TLS / certificate and secret hygiene via **Key Vault, ITSG-33 / PBMM** controls, and self-built monitoring that improved supportability. **Professional Scrum Master I**. Active open-source contributor — **60+ public PRs** including Microsoft’s Azure DevOps tooling (`vsts-team-calendar`, `tfs-cli`) and the **Microsoft Authentication Library .NET** docs.

Professional Experience

Employment and Social Development, Canada Gatineau, QC December 2021 – July 2025

Senior Software Engineer, Infrastructure and Cloud Enablement (BDM Project)

- Enhanced and maintained a **cloud-native, microservices onboarding portal** — **Blazor / C# / .NET front and back end**, Azure SQL, Azure Functions, GraphQL — serving procurement, HR, and IT-ops personnel. Built and extended the **Blazor front end as reusable, component-based UI**; integrated with enterprise IAM via **Azure AD / Entra (OAuth 2.0, OpenID Connect)**; designed and consumed **REST APIs between services** and **published / governed them through Azure API Management** with authentication, throttling, and transformation policies.
- Expanded notification functionality:** Blazor front-end submissions kicked off **event-driven Azure Functions** that created and updated **Azure DevOps work items**, triggered pipelines, provisioned users in on-prem Active Directory (via AAD Connect), and applied group memberships by personnel profile — with bulk submissions fanning out through asynchronous handoffs across the microservices. Built a **scheduled weekly work-item digest emailer** driven by ADO pipelines.
- Leveraged Azure DevOps pipelines end to end** — build, automated test, **Docker** containerization, **Kubernetes** deployment, and **Azure Data Factory** (ETL) orchestration. Administered Azure DevOps for all BDM teams; advised on process templates, branching strategies, and PR workflows; connected **Power BI to ADO Analytics**; ensured **WCAG** accessibility compliance.
- Security, reliability, and supportability:** migrated the portal’s downstream identity from rotating secrets (Key Vault + ADO PATs on 90-day rotation) to **Azure Managed Identities** — eliminating the secret-rotation incident class and properly attributing automated activity. Implemented **ITSG-33 (NIST-based) controls** — TLS / certificate management, **Key Vault** secret hygiene — for SA&A submissions against a **PBMM** profile.

- Designed and shipped **custom ADO web extensions in React, TypeScript, and Node.js** — a build/deploy analytics widget rendering success-rate charts via the ADO Builds REST API, a people-by-project reporter, and the digest emailer above. **Documented developed applications and automated processes**, and ran weekly **demos and design walkthroughs** sharing technical knowledge with junior developers.

Department of Justice, Canada Ottawa, ON

October 2006 – August 2019 (multiple roles)

Full-Stack / .NET Developer · Development Lead · SDET — Legal Case Management System (iCase)

Core contributor and later **Development Lead** on the department's flagship **Legal Case Management System (iCase)** — a distributed multi-tier **ASP.NET / C#** enterprise **web portal** used by **5,000+ government lawyers and external agent firms** across Canada to manage case files, timekeeping, billing, and reporting. Multi-tier with role- and object-based security, integrated with records-management, financial, and email systems, 24/7 high availability.

Development Lead — iCase / Dynamics CRM Modernization (2016–2018)

- **Dev lead** on the live iCase portal while concurrently building its **Microsoft Dynamics CRM** successor. Modified schema and **front-end UI to surface migration status in real time**; engineered Dynamics CRM **plug-ins, custom actions, and workflows** in **C#**; used SSIS for ETL between systems. **Established code-review processes gated on work items** and mentored junior developers.

Full-Stack / .NET Developer (2009–2014)

- Led design and largely solo implementation of the **“My iCase” personalization portal** — the new, **customizable widget-based entry point** to iCase. Ran JAD sessions, produced prototypes, and authored BDD/TDD documentation, then built it on **ASP.NET Web Parts and the Personalization framework with a custom SQL personalization provider, Knockout.js MVVM** component binding (a direct ancestor of the **React / Blazor component** patterns I use today), and async AJAX web-service calls. **Directly analogous to the role's personalization and reusable-component work.**
- **Performance and reliability engineering**: diagnosed a homepage report that took ~10 s uncached — profiled with SQL Profiler and execution plans, then replaced live joins with a nightly-populated flat table (~**50x faster uncached**). Built a **self-initiated availability-monitoring suite** that emailed failures to the right operations group, eliminating volunteer evening/weekend checks. Diagnosed and resolved a **COM Interop memory leak** across the document servers.

Team Lead / Technical Lead — IRCC (2018–2019)

- Led **Agile / Scrum** sprint planning, review, retrospective, daily stand-up, and backlog refinement; introduced poker-planning estimation and capacity forecasting. Implemented custom web extensions in **React / TypeScript / Node.js**; mentored developers in **C# / .NET** and **SQL Server query optimization**; administered **Azure DevOps Server** and its customized process templates.

Software Development Engineer in Test (2008–2009)

- Voluntarily architected a framework in **IBM Rational Functional Tester** that parsed English-language scenario documents and auto-generated executable tests — replacing a ~20–25-tester, two-week manual regression cycle with one person reviewing automated results in five days. **~98% reduction in release-validation effort; ~\$160K saved per cycle.** Trained QA staff and authored the supporting knowledge base.

Employment and Social Development, Canada Gatineau, QC

August 2019 – November 2021

Development Lead, Testing Services

- Co-architected an **internal C# / .NET web application** letting QA staff and business analysts author structured test scenarios via a web UI; designed the **SQL Server** schema and **co-designed the REST API** for bidirectional integration with **Azure DevOps pipelines** — versioned script-fetch endpoints and results-push endpoints.

- Engineered Azure DevOps **CI/CD pipelines for automated testing** — provisioned agents to run **parallel Selenium UI tests** against deployed applications and integrated results back into work items. Code review and standards enforcement for the team.
- Championed the departmental transition from **LoadRunner to JMeter**, cutting licensing costs while maintaining robust performance testing.

Department of Citizenship and Immigration, Canada Ottawa, ON April 2014 – December 2015

Senior Software Engineer · TFS / Release Engineering Lead

- Hired alongside one other engineer to take over a release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop apps** and replaced them with lightweight **PowerShell** calling **REST APIs** — most contractor tooling was simply deleted, and **everything ended up in source control**. Bias for action and ownership.
- Implemented and deployed **web extensions in React / TypeScript / Node.js**; maintained a fork of Microsoft's `vsts-team-calendar` extension with configurable colours, Outlook integration, and release-management features, contributed back upstream. **Installed, configured, customized, and upgraded** the ADO/TFS platform across dev / UAT / production.

Abrams Software Kingston / Toronto, Ontario (Remote)

August 2025 – Present

Full-Stack Developer · Independent

- **Chess Attack Visualization — React 19 + TypeScript** training app (Vite, Vitest / Jest) consuming the **Lichess.org REST API**, with a **NestJS** auth proxy implementing the **OAuth2 PKCE** flow. Diagnosed a missing feature in `chess.js`, opened [upstream PR #572](#), and published a fork on npm. Live at [paulabrams.ca/chess-attack-visualization](#).
- **TWNR** — architected and shipped a **public-facing B2C multiplayer game as a full-stack web application, solo**: Dockerized **TypeScript** monorepo, **Node.js (Express) + WebSocket** server backed by **PostgreSQL** with per-player **JWT auth** and a **JSON-Schema-generated REST API contract**. Live at [twnr.paulabrams.ca](#).
- **LLM code evaluation** for [DataAnnotation.tech](#) and [Stellar.ai](#) — prompt fixtures and grading rubrics across **TypeScript, React, Node / NestJS, .NET / C#, Python, Go, Rust, and SQL**, including **security audits with CWE / CVSS-graded inventories**. **60+ public PRs** to outside repos, including the **Microsoft Authentication Library .NET docs** and [canada-ca/platform-security](#).

Education

Bachelor of Computer Science (Honours) - Carleton University

September 1999 – May 2003

Certifications & Awards

[Professional Scrum Master I](#) (Scrum.org) · [ITIL v3 Foundation](#) (Loyalist Certification Services) · DOJ Team Merit Award 2012; Team Merit Award 2011; Team Spirit Award 2009

Technical Skills

.NET / C#: **.NET 6 / 8 / .NET Core, C#, Blazor**, ASP.NET / ASP.NET MVC, ADO.NET, Entity Framework, LINQ · component-based UI, reusable components

Front-End / UI: **Blazor, React 19, TypeScript**, JavaScript, Vite, Knockout.js MVVM, HTML, CSS, JSON, AJAX · **responsive design, WCAG / AODA accessibility**, design-to-component translation (Figma-style design handoff)

Azure DevOps: **work-item analysis & delivery, ADO pipelines (YAML) for automated build / test / deploy**, ADO REST API, custom web extensions, ADO Server administration & process-template customization, Power BI on ADO Analytics

Cloud / Azure: App Service, **Azure Functions** (event-driven), SQL, **API Management, Key Vault**, Storage, **Managed Identities, Entra / AAD**, Data Factory · **Docker, Kubernetes**

Server-Side / APIs: REST API design, GraphQL, WebSocket · NestJS, Node.js, Express · n-tier client/server, **cloud-native microservices**, event-driven workflows

Security: OAuth 2.0 / OpenID Connect / PKCE, Entra / AAD IAM, TLS / certificate management, Key Vault secret hygiene, Managed Identities, **ITSG-33 / PBMM controls**, CWE / CVSS-graded audits

Databases: MS SQL Server **2005 → 2019** (T-SQL, stored procedures, indexing, execution-plan tuning), Azure SQL, PostgreSQL

Testing: Selenium, Playwright, **Jest, Vitest**, xUnit / NUnit, IBM Rational Functional Tester, JMeter

Process / SDLC: Agile / Scrum (PSM I) — daily stand-ups, sprint planning & review, simplified SCRUM · code review, mentoring, technical documentation & knowledge-sharing · ITIL v3 Foundation